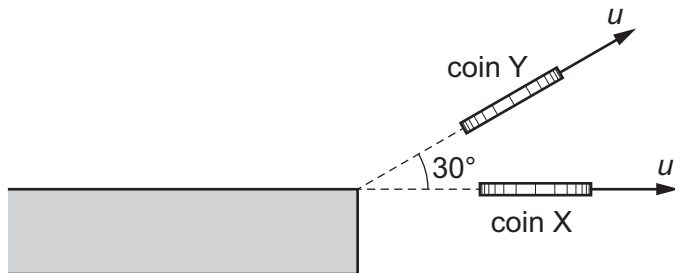


- 7 Two coins are projected from a horizontal table at the same initial speed u .

Coin X is projected horizontally.

Coin Y is projected upwards at an angle of 30° to the horizontal.



Both coins hit the horizontal ground without bouncing.

Assume the air resistance on each coin is negligible.

Which statement about the motion of the coins is correct?

- A Both coins hit the ground with the same speed.
- B Both coins travel the same vertical distance.
- C Coin Y hits the ground before coin X.
- D Coin Y has a smaller vertical acceleration.

- 8 Two spheres are released from rest at equal heights above the ground